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Outcomes of Kidney Transplantation: Identification, Measures, and Evaluation

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INTRODUCTION

Chronic kidney disease (CKD) is a global public health problem that imposes a substantial burden on patients, families, and healthcare systems.[1] Kidney transplantation provides superior survival compared with dialysis but does not fully correct quality of life (QOL) and survival[2,3] compared with the general population. The overall risk of death in transplant recipients is at least three times higher than the general population, and transplant recipients have significantly reduced QOL, even compared with patients with other chronic disease. Strategies such as optimization of matching, crossmatching, and surgical techniques; improvement in perioperative care; effective antiviral, antifungal, and antibacterial prophylaxis; and contemporary immunosuppression have led to improvement in patient and allograft outcomes over the past two decades. Improvement in short-term outcomes is accompanied by evidence from the United Kingdom, Australia, and Europe on gains in long-term graft survival. Data from the Collaborative Transplant Study have shown that between 1985 and 1996, the 1- and 10 year-graft loss hazard rates have declined by 64% (95% confidence interval [CI], 61%–66%) and 45% (39%–50%), respectively. Between 2000 to 2015, the reduction in graft loss hazard rates at 1 and 10 years were 22% (12%–30%) and 64% (45%–76%), respectively.[4]

Despite these improvements, there are important challenges in improving the health and overall well-being of transplant recipients. Current immunosuppression regimens cannot completely prevent rejection and graft loss and can lead to undesirable complications such as cancer and infection. Importantly, the outcomes that researchers study and clinicians target often do not align with those that patients prioritize.[5] Patient engagement is a critical element of clinical research that can enhance the credibility of the findings and improve their translation into practice.

This chapter describes outcomes after transplantation including graft health, mortality, and long-term complications of immunosuppression and includes a brief description of the tools used to assess these outcomes. The chapter explicitly focuses on outcomes that are prioritized by patients and families, especially those that are not frequently considered by clinicians and health professionals, such as life participation.

GRAFT HEALTH AND PATIENT SURVIVAL

Graft health is a broad umbrella term for direct outcomes of the kidney transplant. Important aspects of graft health include patient survival, short- and long-term graft survival, graft rejection, and graft function.

Definitions that characterize patient and allograft survival include the following:

- *Patient survival* refers to the overall survival (OS) of the transplanted person measured from the time of transplantation to the time of death; censor at the end of follow-up time or the time the person was lost to follow.
- *Death with a functioning graft* is a major cause of graft loss after kidney transplantation. This occurs when a patient has sufficient graft function and does not need either retransplantation or dialysis at the time of death. The date of death is recorded as the last follow-up visit with function.
- *Graft survival* is the absence of irreversible graft failure, meaning the requirement for retransplantation or long-term dialysis therapy. This can be further divided:
  - *Death-censored graft survival* is time from date of transplantation to the occurrence of irreversible graft loss; censor on the date at which the patient was lost to follow-up, died, or at the end of follow-up time.
  - *Overall graft survival* is time from transplant to graft loss and/or death; censor on the date at which the patient was lost to follow-up or end of follow-up time for recipients with a functioning graft.

Measures of patient or graft survival can be reported with or without adjustment for potential confounders and typically at a specified time point (such as 30-day, 1-year, 5-year, or 10-year). Transplant data registries, such as the International Registry in Organ Donation (IRODaT), Organ Procurement and Transplantation Network (OPTN), United Network for Organ Sharing (UNOS), Scientific Registry of Transplant Recipients (SRTR), United States Renal Data System (USRDS), European Dialysis and Transplant Association Registry (ERA-EDTA), Collaborative Transplant Study (CTS), and the Australia and New Zealand Dialysis and Transplant Registry (ANZDATA), provide important data regarding the country-specific graft and patient survival (both short- and long-term) over the past 50 years.

Early Posttransplant Outcomes

There are high upfront risks associated with kidney transplantation, driven by surgical complications and adverse effects of intense immunosuppression early in the transplant course. A longitudinal, observational study of over 200,000 patients on long-term dialysis in the United States[6] showed that patients who were listed for transplantation (46,164) and received a first deceased-donor transplant (23,275) experienced a 2.8-fold higher risk of death in the first 2 weeks after the transplant compared with those who remained on the waitlist. The risk of death became similar between transplanted and waitlisted patients around 106 days, and a net survival benefit was seen after 244 days posttransplantation.[6] Importantly, the study showed a substantially lower annual death rate for those transplanted (3.8 deaths per 100 patient-years) compared with those who remained waitlisted on dialysis (6.3 per 100 patient years) and on dialysis but not suitable for waitlisting (16.1 per 100 patient years), with benefits particularly favoring young patients (20–39 years old) and young diabetic patients.

Long-Term Posttransplant Outcomes

Interpretation of longer-term outcomes needs careful consideration of the case mix (such as recipient comorbidities, organ quality, immunologic risks, donor-recipient matching), transplant era (evolution of immunosuppression protocols and practices), and country-specific and resource-related factors (such as the impact of socioeconomic factors, access to care, and affordability of medications, which may vary between countries; Fig. 114.1). These factors may have accounted for the between-country differences in the adjusted graft loss rates in the United States, United Kingdom, and Australia and New Zealand, with the risk of graft loss at 1 year posttransplant being highest in the United Kingdom (Hazard ratio [HR] 1.22), followed by the United States from year 2 onward. The risk of allograft loss remains the lowest in Australia across all eras.[7] Other studies have reported a wide range of 10-year patient survival, ranging from 34% to 78% for transplants performed after the year 2000.[8–11] The impact of regional and ethnic differences was again highlighted in a study comparing European versus North American kidney transplant recipients from the UNOS and CTS for Europe databases. The 10-year graft survival rates between 2005 to 2008 were 56% in Europe, and 46%, 48%, and 34% for White, Hispanic, and Black Americans, respectively.[10]

Data from the Netherlands Organ Transplant Registry suggested that deceased-donor criteria (brain vs. circulatory death) were not a predictor for patient- or death-censored graft survival.[9] However, donor comorbid diseases, such as type 2 diabetes and hypertension, were risk factors for allograft survival.[12] For transplants performed in 2014 to 2015, the death-censored graft survival rates at 1 and 5 years were 98% and 83%, respectively, for deceased-donor grafts, and 99% and 91% for living-donor kidney grafts in Australia[13] (Fig. 114.2A). The rate of graft failure (censored for death) was approximately 3 per 100 graft-years, with chronic allograft injury being the dominant cause of graft loss[13] (Fig. 114.3).

Patient OS after transplantation differs by era and recipient age. The 5- and 15 year patient survivals after primary living-donor kidney transplant varied from 89% in the 1990–94 period to 94% in the 2000–04 period and 75% in the 1990–94 period to 80% in 2000–04. For primary deceased-donor kidney transplants, the 5- and 15- year survivals were 84% in 1990–94 and 89% in 2004–09 and 53% in 1990–94 and 64% in 2000–04, respectively (see Fig. 114.2B). The reported 5-year graft patient survival for older kidney transplant recipients (≥65 years old) transplanted between 2006 to 2016 was 79%.[14] Older transplant recipients also incur a survival advantage compared with age- and

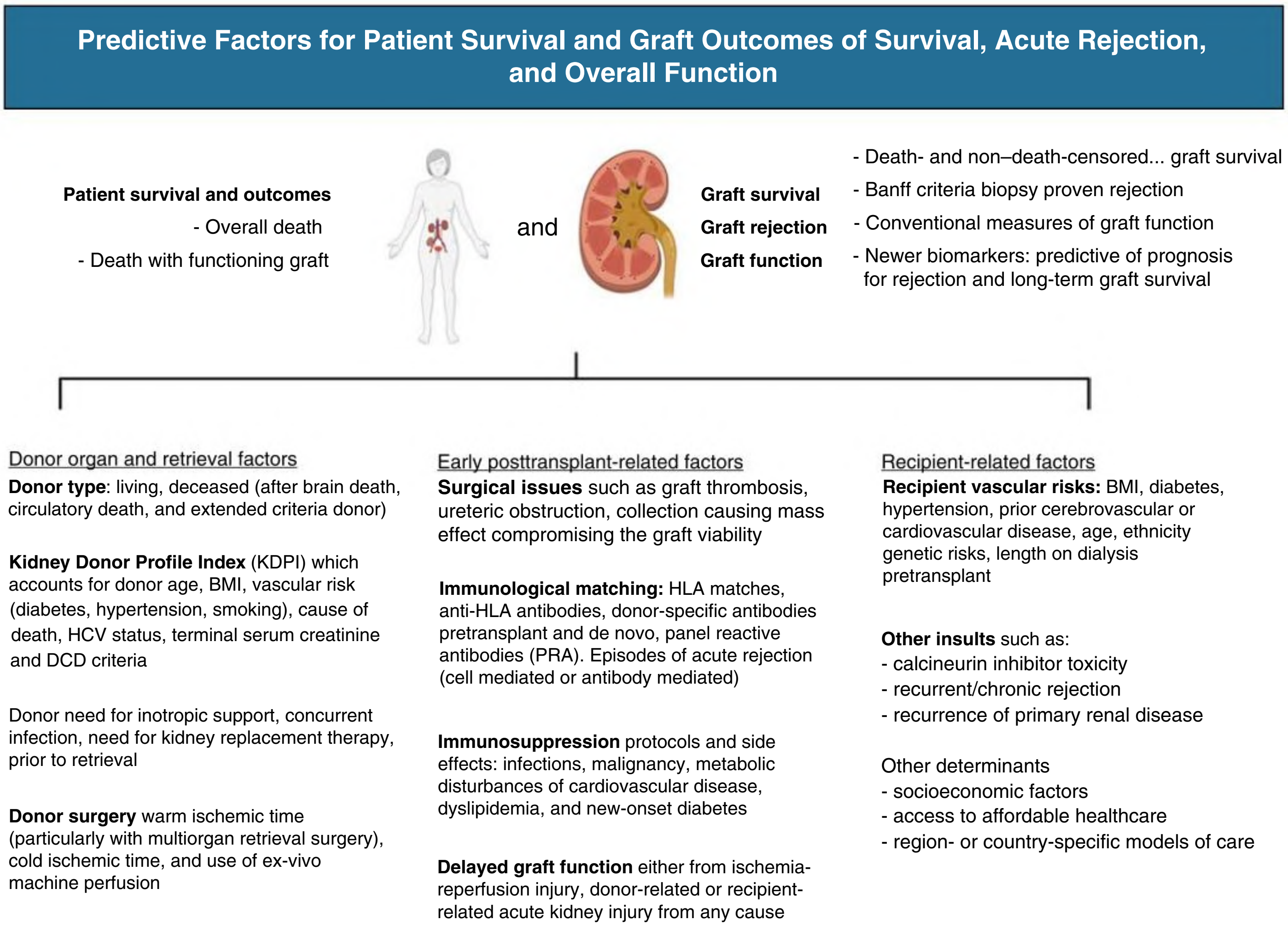

Fig. 114.1 Predictive factors for patient survival and graft outcomes of survival, acute rejection, and overall function. *BMI*, Body mass index; *DCD*, donation after cardiac death; *HCV*, hepatitis C virus; *HLA*, human leukocyte antigen. (Created using Biorender.com.)

Graft and Patient Survival for Deceased and Living Donor Kidney Transplantation Across Eras in Australia

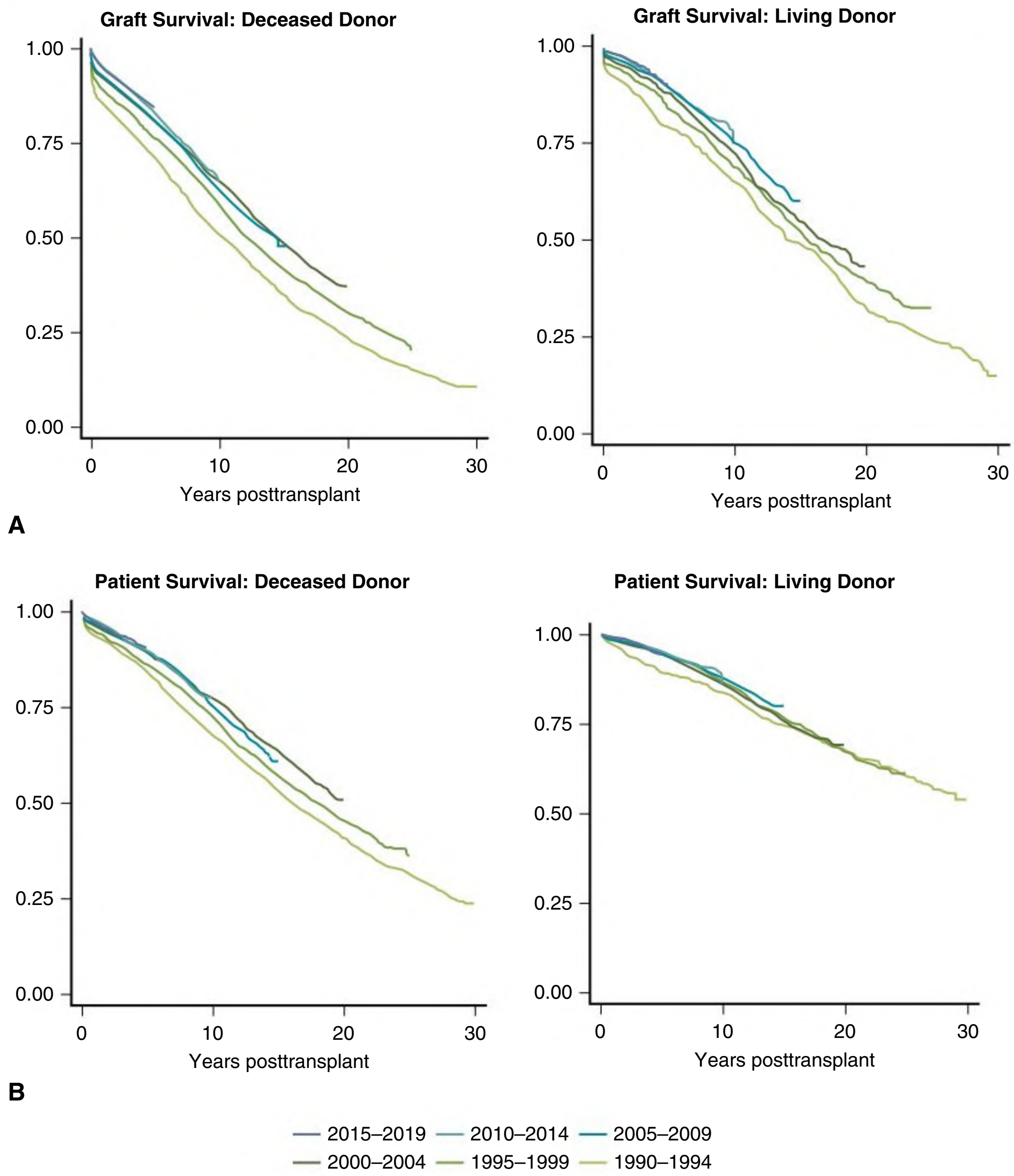

Fig. 114.2 Graft and Patient Survival for Deceased and Living Donor Kidney Transplantation Across Eras in Australia. (A) Primary graft survival. (B) Patient survival. (A, From ANZDATA Registry Annual Report 2020.)

sex-matched patients who remain on the waiting list,[15] but OS remains inferior compared with younger counterparts.[16] Donor age is also a major factor that influences transplant outcomes. Younger recipients of older donor grafts are at risk for poorer long-term graft survival.[16–18]

Acute Rejection Episodes After Transplantation

Acute rejection is an important outcome for patients and health professionals. Diagnosis of acute rejection is based on biopsy scoring using the Banff criteria for rejection.[19,20] Studies typically report on biopsy-proven acute rejection (BPAR)—whether cell-mediated or antibody-mediated rejection (ABMR)—to satisfy the predefined Banff criteria. The Banff classification was first developed in 1991 and has had several iterations over the past 30 years,[20,21] including updates as recently as 2019. Interpretation of studies require careful consideration given that the classification is typically-updated every 2 years, the addition of ABMR criteria after the 1997 meeting, and the acute versus chronic ABMR classification detailed in the 2005 update. Studies over the past 30 years have subtle variations in the definitions of BPAR, and definitions may be improved by incorporating diagnostic, predictive, and/or prognostic biomarkers (select examples listed in Table 114.1) and more precise definition of phenotypes of acute rejection defined by molecular and genomic markers.[22–26]

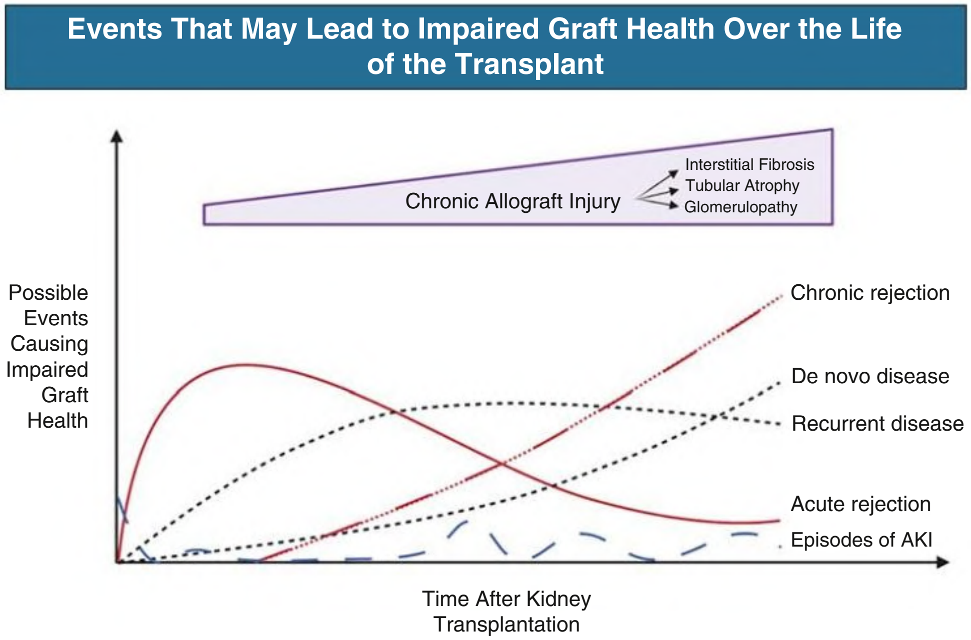

Fig. 114.3 Representative Events That May Lead to Impaired Graft Health Over the Lifetime of the Transplant. Early posttransplantation, acute rejection, and delayed graft function pose the highest risk to graft function, which reduce over time and are replaced by events such as chronic rejection and recurrent or de novo kidney disease superimposed by episodes of acute kidney injury (AKI) from any cause. (Created using Biorender.com.)

TABLE 114.1 Select Noninvasive Biomarkers for Solid-Organ Transplant Rejection

Marker	Source	Reference
Donor-specific antibodies	Blood	76
Molecular biomarkers group or panels	Blood, urine, and/or kidney biopsy Examples: kSORT, CTOT-8, GoCAR studies	77–79
T-cell immunoglobulin and mucin-domain containing-3 (TIM-3)	Urine	80
Chemokine (C-X-C motif) ligand 9 (CXCL9)	Urine	81, 82
Chemokine (C-X-C motif) ligand 10 (CXCL10)	Urine	81
Cell-free DNA	Blood	83
Granzyme B, perforin and Fas-ligand	Blood	84
Forkhead box P3 (Foxp3)	Blood and urine	85
OX40, (CD134) OX40, ligand, programmed cell death protein 1 (PD-1), PD-2, FOXP3	Urine	86
MicroRNA species	Biopsy, blood, or urine	87–90

Acute rejection occurred in approximately 18% and 20% of patients with first and subsequent allografts within the first 6 months after deceased and living-donor kidney transplants. Early rejection (within the first 6 months posttransplantation) adversely affects longer-term patient and graft outcomes and is associated with an increased risk of death with a functioning graft, death because of cardiovascular disease (CVD), or cancer.[11]

Kidney Transplant Function

Kidney function is an important outcome that is usually reported using surrogate markers of serum creatinine (SCr) or estimated glomerular filtration rate (eGFR). These outcomes have known limitations, particularly with variations with sex, race, muscle mass, and age; aberrant results when drugs or conditions impair the tubular secretion component; and the relative insensitivity to small changes in function because of the reciprocal relationship between SCr and GFR. There are several methods for estimating eGFR, most commonly the Modification of Diet in Renal Disease (MDRD) and Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equations (see Chapter 3).[27,28] Despite their limitations, recording the eGFR and observing the magnitude of eGFR decline, slope, or drop of at least 30% in function is useful for predicting patient and graft survival.[29] The GFR slope and percentage decline in GFR are also used to detect the progression of kidney disease.[30,31] Proteinuria is an important biomarker in kidney transplantation, as it is in the CKD setting.[32] Proteinuria originating from the allograft (rather than native kidneys) can be from various etiologies (including transplant glomerulopathy, recurrent disease, and mammalian target of rapamycin [mTOR] inhibitor use), but even proteinuria of greater than 500 mg/day is associated with reduced graft survival[33] and development of de novo human leukocyte antigen (HLA) antibodies.[34]

Early Graft Outcomes

Delayed graft function (DGF) is typically defined as the need for dialysis within the first 7 days, but there are 18 definitions used in literature to define this entity.[35,36] The reported incidence of DGF varies from 5% to 50% depending on the definition and can be associated with increased risk of rejection[37,38] and reduced graft survival.[39] Recent evidence also suggested that the severity (based on the duration) of DGF has a "dose effect" on death-censored graft loss.[40] For patients with "slow graft

TABLE 114.2 Select Novel Biomarkers for Acute Kidney Injury

Marker	Source	Reference
Neutrophil-gelatinase-associated lipocalin (NGAL)	Blood and urine	91, 92
Interleukin-18	Blood and urine	93, 94
Kidney injury molecule-1 (KIM-1)	Urine	95, 96
Cystatin C	Blood	97, 98
Liver-type fatty acid binding protein	Urine	96
YKL-40	Urine	99, 100
Tissue inhibitor of metalloproteinases 2 (TIMP2)	Urine	101
Insulin-like growth factor binding protein 7 (IGFBP7, angiomodulin)	Urine	102
Angiotensinogen	Urine	103
Cell-free DNA	Blood	104

function" (poorer immediate function but not requiring dialysis within the first 72 hours), DGF was associated with worse outcomes in terms of overall graft loss and death-censored graft loss.[41] Similar to acute rejection, the definition of DGF can be precisely defined by incorporating noninvasive biomarkers to the clinical or laboratory definition(s) in the future. Some examples of biomarkers being investigated for acute kidney injury (AKI) in general are presented in Table 114.2.

Long-Term Graft Outcomes

Chronic allograft injury (CAI; previously known as chronic allograft nephropathy) is a histologic description of fibrosis, atrophy, and vascular damage in the kidney. The term interstitial fibrosis/tubular atrophy (IFTA) was introduced by the 2005 Banff Working Group[42] to capture this pathologic appearance, which represents the irreversible and common end pathway of many mechanisms of injury, including acute or chronic rejection, subclinical rejection, long-term calcineurin toxicity, polyoma virus infection, recurrent or de novo glomerulonephritis, or any cause of AKI seen in the general population.[43–45] IFTA is prone to sampling error from the biopsy procedure and ascertainment bias because of variation in the sampling rates of biopsies for indication only or indication plus surveillance biopsies. The quantification of IFTA alone is insufficient to predict long-term graft loss,[46] but inflammation in areas of IFTA (i-IFTA) may be associated with accelerated IFTA, chronic glomerulopathy, reduced kidney function, and long-term death-censored graft loss.[47,48] Thus, IFTA and i-IFTA are useful surrogate markers for kidney transplant outcomes.

The kidney donor risk index (KDRI) is a validated prediction tool for long-term graft survival and includes a numeric representation accounting for donor age, race, terminal creatinine, comorbid hypertension or diabetes, cause of death, donation after cardiac death, height, weight, and HCV status; and transplant factors including HLA-mismatch, cold ischemia time, and en-bloc double kidney transplant.[49,50] A higher KDRI denotes the higher risk of graft failure or a "poorer" quality organ but this is not absolute, with similar graft outcomes reported for KDRI between 35 to 85 and KDRI greater than 85.[51]

In recent years, the search has intensified for predictive or prognostic blood, biopsy, and/or urinary biomarkers that correlate with long-term outcomes in kidney transplantation, particularly those using proteinomics and genomic scoring sets.[24,52–54] Although there has been acceleration in molecular technologies and bioinformatics analysis, these newer research tools are yet to be translated into mainstream clinical practice and are not further discussed here.

COMPLICATIONS: MALIGNANCIES AND INFECTIONS

Cancer and infection are the two most feared complications of immunosuppression and are thus key research priorities for patients and clinicians.[5] Unlike CVD, for which outcomes have improved considerably over time, the risks of developing atypical infections and cancers remain high.[55]

MALIGNANCY

The overall risk of cancer is increased at least twofold to threefold compared with the age- and sex-matched general population, with the greatest increased risk from viral-related cancer. However, the risk of developing breast and prostate cancer is comparable to the general population.[56] The overall incidence of de novo cancer among kidney transplant recipients varies between 1000 and 1300 per 100,000 person years, and the overall risk increases with age. The cumulative incidence of de novo cancer (including nonmelanoma skin cancer) at 10 years after kidney transplantation is around 40%, increasing to 60% at 20 years. Once cancer has developed, the risk of death attributed to cancer is threefold higher than for patients with cancer in the general population.[57]

Measuring Cancer Outcomes in Kidney Transplant Recipients

Epidemiologic data such as incidence and mortality of cancer in transplant recipients are best obtained from national cancer-based registries. Linkage of data from different sources (e.g., linking data from a cancer registry to data from a kidney transplant registry) may allow more detailed analyses that cannot be done with either alone. Cancer incidence and mortality rates are presented by categories of cancer sites and types in kidney and general population cancer registries. Real-time data, including incidence rates (sex- and age-specific) and the cumulative cancer incidence (by cancer site), may be routinely provided by the registries to transplant units as a quality assurance measure. If historical incidence data are available, with sufficient and accurate number of cases and the denominator, recorded incidence rates can be projected to provide longer-term estimation within the population of interests using prediction modeling techniques.[58]

Although unadjusted incidence rates are useful to describe how many kidney transplant recipients develop cancer or die of cancer within a certain time period, they do not account for the age, sex distribution, and patient characteristics in the population of interest. Standardization gives a single figure to compare an incidence rate in two populations and accounts for the different population distributions.

Relative Mortality

Obtaining accurate information on the cause of death is notoriously difficult, especially using data from death certificates. To overcome some of these challenges, the risk of cancer-related death in kidney transplant populations can be assessed by assessing relative cancer survival. Relative survival can be defined as the ratio of kidney transplant recipients with cancers that have survived within a certain time period to the proportion of expected cancer survivors in a comparable cohort in the general population. Alternately, the comparator could be a similar cohort (age- and sex-matched) of patients without cancer in the general population. However, this method assumes that cancer deaths are independent of the other competing causes of death.

Competing Risk Analyses

Accounting for the competing risk of other causes of death is potentially important when assessing the risk of death from cancer in kidney transplant recipients. In survival analysis, censored observations (participants who complete follow-up without experiencing the outcome of interest) contribute to the total number of participants at risk until the end of their follow-up. These analyses assume that factors that lead to censoring are independent of those that lead to the outcome of interest. However, if the outcome of interest is precluded by another event, the "independence" assumption is violated. For example, a transplant recipient who dies of infection cannot experience another death from cancer and is no longer "at-risk" for experiencing another type of death. Specific techniques such as the Fine and Gray method can be used to generate the subdistributional hazard ratios and estimate the risk of cancer-related death, assuming the recipient had not died of infection. Ignoring competing risk can lead to misleading results. For example, some analyses have suggested that patients with diabetes are protected from cancer compared with those without diabetes; the true explanation is that (compared with people without diabetes) these patients are at higher risk of early death from other causes and thus are less likely to live long enough to die of cancer.

Management of Cancer After Transplantation

Partnership between transplant professionals and oncologists is the key to cancer management in complex kidney transplant recipients. Judicious reduction in immunosuppression, particularly antiproliferative agents and those that affect T cells, may induce remission of certain cancer types, such as posttransplantation lymphoproliferative disorder (PTLD) and Kaposi sarcomas. However, this approach needs to be balanced carefully with the risk of allograft rejection. There are emerging studies to suggest that mTOR inhibitors (mTORIs) may reduce the risk of method can be used (particularly squamous cell carcinoma [SCC]) and Kaposi sarcomas.[59] However, the benefits of mTORIs in recipients with solid organ cancers are uncertain and mTORIs are poorly tolerated, with a discontinuation rate of more than 30% in trials.[60] There is also an increased risk of cancer and all-cause death among recipients who received mTORI as standard immunosuppression compared with those maintained on calcineurin inhibitors.

New targeted anticancer therapies, including checkpoint inhibitors and other immunotherapies, are now available to treat advanced-stage solid organ and hematologic malignancies, but most trials assessing these new agents have excluded transplant recipients. There are case reports and series suggesting that the use of anti-PD1, cytotoxic T-lymphocyte–associated antigen 4 (CTLA-4), and other immune modulators such as lenalidomide and pomalidomide in transplant recipients can lead to acute allograft rejection.[61] A recent observational multicenter study on the safety and efficacy of immune-checkpoint inhibitors (ICI) indicated approximately 40% of the cohort who received ICI developed acute rejection, and this compared with 5.9% in the non-ICI groups. Of those who experienced acute rejection, about 65% experienced graft loss. The median time from ICI initiation to acute rejection was 24 days. Key factors that may reduce the risk of acute rejection were mTORI use at the time of ICI initiation, and triple agent immunosuppression. The median survival in patients with metastatic SCC was 19.8 months in the ICI cohort, compared with 10.6 months in the non-ICI arm.[62]

INFECTIONS

Infections are major causes of early posttransplant death (within the first 12 months posttransplant), particularly in low- to middle-income countries where prophylactic treatments for viral and bacterial infections are not universally affordable. The timing, severity, and etiology of the infections are dependent on the recipients' epidemiologic infectious exposures and the individual's state of immunosuppression[63] (see Chapter 110). The types of infection also vary depending on the time since transplantation, with the risk of acquiring nosocomial infections being the highest during the first weeks after transplantation, followed by opportunistic and community acquired infections thereafter.[63] Unlike cancer, detailed data on the timing, frequency, and burden of infectious disease in transplant recipients are generally not available from national registries. Details about the incidence, prevalence, and exposure to certain types of infections and the responses to screening, preventive, and treatment strategies are often limited to single-center sources.

Prevention and Prophylaxis

Effective prophylactic and preventive strategies are available for a range of infections, including cytomegalovirus (CMV), human papillomaviruses (HPV), and pneumocystis pneumonia infections. Options for CMV prophylaxis include valganciclovir, oral ganciclovir, intravenous (IV) ganciclovir, or high-dose oral valacyclovir, corrected for kidney function.[64] For HPV, quadrivalent vaccines (against genotype 6, 11, 16, and 18), and, more recently, the HPV 9-valent vaccines (against 5 additional genotypes of 31, 33, 45, 52, and 58), are highly effective and have an overall efficacy of 99% to 100% for the prevention of cervical intraepithelial neoplasia.[65,66] HPV vaccination is indicated in both males and females aged 9 to 25 years in the general population for the prevention of HPV-related malignancies. Some recent data have shown that vaccination is also efficacious in women up to age 45 years. In the transplant population, HPV vaccines are generally safe, but evidence for decision making is limited to observational analyses. In a single cohort study, seropositivity was only detected in approximately 50% to 60% of patients, depending on genotypes, and higher tacrolimus levels were also detected in nonresponders.[67]

Early Detection

Although prevention is the key to preventing serious illnesses in the transplant populations, early detection of disease in patients at risk may provide the opportunity to initiate effective treatments and prevent advanced-stage disease. Polyomavirus is an important pathogen associated with interstitial nephritis and nephropathy in kidney transplant recipients. The risk of graft loss, once nephropathy has developed, ranges from 30% to 50% within 5 years.[68] In the absence of effective antiviral treatments, management of polyomavirus (BK) nephropathy relies on early detection and timely reduction of immunosuppression with urine real-time polymerase chain reaction (RT-PCR) or urine cytology. Additional studies of diagnostic test accuracy are required to determine which test or tests best distinguish a transplant recipient with BK nephropathy from those without the disease. The conduct and reporting of all diagnostic test accuracy studies should adhere to the Standards for the Reporting of Diagnostic Accuracy Studies (STARD) guidelines.[69]

Economic Evaluations

Economic evaluation can be used to assess the costs and benefits of interventions and help policy makers identify interventions that maximize health benefits and minimize resource use.[70,71]

A recent analysis reported that living-donor kidney transplant is cost-saving compared with dialysis because most recipients live longer and incur fewer costs. For those who received a lower kidney donor profile index (KDPI) deceased-donor kidney transplant, the incremental cost-effectiveness ratio (ICER) ranged between (USD) $36,000 to $40,000 per quality-adjusted life-year (QALY), and approximately $60,000 per QALY for higher KDPI deceased-donor kidney transplant and in patients who are immunologically complex. Despite the higher costs, the survival advantage of kidney transplantation over dialysis is so substantial that all options are highly cost-effective, based on a willingness-to-pay threshold of $100,000 (USD) per QALY.[72]

Although the cost-effectiveness and cost utility of kidney transplantation compared with dialysis are not in doubt, more uncertainty remains about which new treatments will lead to further gains, compared with current management. In some cases, clinical trial data are not available to guide such decisions, and in this circumstance an economic evaluation could be used to estimate the likely costs and effects of any interventions. A published example is the evaluation of cidofovir in kidney transplant recipients with BK nephropathy. Cidofovir is often used in patients with significant nephropathy resulting from BK infections. All efficacy and adverse effects data informing cidofovir use in transplant recipients are from observational studies. Decision analysis suggests that the addition of cidofovir may incur a small survival gains, approximately 2.2 days with savings of approximately $20,000 AUD over the lifetime of a recipient. However, uncertainties exist because of unclear clinical efficacy data, where the incremental health benefits could range from –1 (harm rather than benefit) to 1.3 life-years-gained, with incremental costs varying between an extra $30,000 to savings of $27,000. Other examples of how economic analysis has been used to assess the likely impact of treatment strategies on outcomes in kidney transplantation include immunosuppression use, induction therapy, cancer screening, and machine perfusion.

PATIENT-REPORTED OUTCOMES AFTER TRANSPLANTATION

Kidney transplantation is intended to improve both the survival and QOL of patients with kidney failure, but the adverse effects from immunosuppression and coexisting comorbidities such as infections, vascular disease, and mental health disorders may affect the recipients' capacity to return to normality, participate, and enjoy daily life activities such as work, education, and recreation. Other factors including age, sex, the types of transplant received, and the duration since transplantation also have a major impact on the overall QOL experienced. People adapt to their illnesses, but the extent and the time needed for the adaptation are associated with the types of transplants they receive and the complications that occur during the perioperative and postoperative periods. Compared with deceased-donor transplant recipients, patients who have received a living-donor transplant experienced better social functioning, vitality, and physical functioning in the first 5 years after transplantation but not thereafter.[73]

Inclusion of patient-centered outcome data in research and clinical care allows for explicit evaluation of the end-user engagement, satisfaction, and potentially real-world adherence. There are several reasons for incorporating patient-reported outcomes in the care of transplant recipients. First, patients are accurate in describing their own symptoms, pain, functions, and QOL. Second, patient-reported outcomes can be used to support shared decision making and patient-centered care. Finally, patient-reported outcomes can also generate valuable data on the treatment effectiveness of interventions, the side effects, and delivery of care in clinical practice. The Standardised Outcomes in Nephrology–Kidney Transplantation (SONG-Transplant) Initiative has identified a core set of clinical trial outcomes that reflect the priorities of key stakeholders,[74] based on a systematic review and an international Delphi survey that included key stakeholders such as patients, caregivers, and health professionals. The full range of outcomes are shown in Fig. 114.4. Although there was general agreement between healthcare professionals, patients, and caregivers that graft survival, function, cancer and infections, and QOL should be included in the core outcomes set, there was incongruency in their opinions regarding the best tools to measure these domains.

Outcome Measures

Life participation is an example of an outcome that is critically important for transplant recipients and their families but is not much considered by clinicians or researchers. Life participation can be defined as the ability to participate in activities that provide a sense of fulfillment, enjoyment, control, and hope. This encompasses a range of activities including, but not limited to, paid and volunteer work, family duties, social functions, recreational and leisure activities, and hobbies. Although life participation is a high priority for patients and

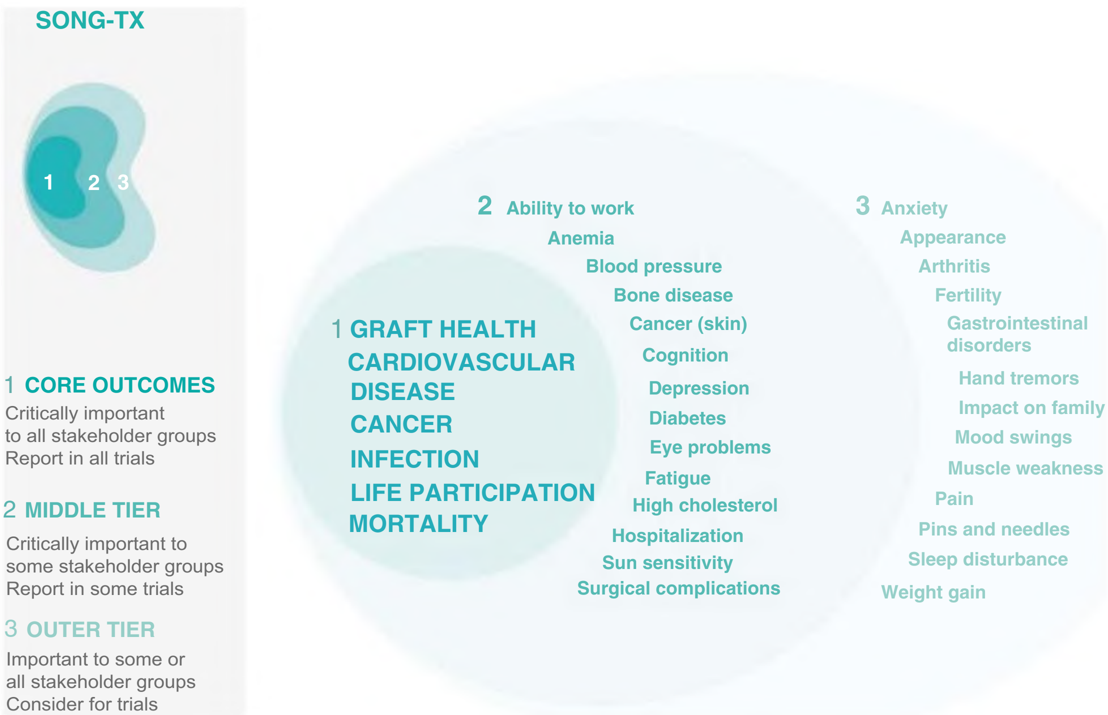

Fig. 114.4 Standardised Outcomes in Nephrology (SONG)-Transplant Core Outcomes. (Courtesy Outcome Measures in Rheumatology [OMERACT]).

their families, the domains, psychometric properties, and content of the outcome measures for life participation in transplant recipients have not been defined.[75] More importantly, this outcome is infrequently explored and inconsistently evaluated in intervention trials in transplant recipients. Life participation can be conceptualized into two dimensions: obligatory and nonobligatory. Obligatory activities are crucial for survival and independence such as work, education, and household chores, whereas nonobligatory events include mainly leisure activities but are important to maintain the well-being of the individual and return a sense of normalcy. There are many tools that measure life participation. Most of these evaluated constructs (such as QOL of the patients and caregivers) included life participation in the questionnaire. Few specifically evaluate the specific aspects of life participation such as daily activities of living and disability assessment. However, many of these measures have not yet been validated in kidney transplant populations, and further research is needed.

CONCLUSION

Kidney transplant recipients value optimal graft function and survival. Although most patients are willing to accept some adverse side effects as being necessary to avoid complications such as acute rejection, the high incidence of other undesirable outcomes such as infection and cancer indicate that further research is needed. Finally, the relative lack of data on how best to improve other patient-important outcomes such as life participation and other symptom burden should be a research priority for the transplant community.

SELF-ASSESSMENT QUESTIONS

1. Actuarial survival is a:
   A. method of calculating how long a patient will live after kidney transplant failure.
   B. statistical method of maximizing the information available for analysis of outcome from a group of patients and events.
   C. way of making "actual results" seem more plausible.
   D. statistical method to estimate the outcomes of patients who are lost to follow-up.
2. The composite endpoint accepted by the US Food and Drug Administration for immunosuppressive drug trials in transplantation includes:
   A. graft failure and patient death.
   B. graft failure, chronic rejection, and lost to follow-up.
   C. acute rejection, graft failure, lost to follow-up, and patient death.
   D. biopsy-proven acute cellular and acute antibody-mediated rejection.
3. Meta-analysis is a:
   A. way of deciding which of several randomized trials is correct.
   B. statistical software package.
   C. method of combining the results from all similar trials to gain statistical power.
   D. trial design used in large studies to decide which factor is most important in determining an outcome.

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